

StudyShare

Technical Specification & Review

A comprehensive architectural analysis, detailed database schema map, technical workflow overview, and review of the newly implemented Lecturer Subject Assignment System extension.

Project Details & Specifications

Version:	2.0 (Stable Release)
Date of Issue:	May 29, 2026
Target Audience:	Admin, Head of Departments, Developers, Lecturers
Primary Stack:	PHP 8.2 (XAMPP), MySQL, Vanilla CSS, Tailwind CSS
API Integrations:	Google Drive API v3, Google OAuth2, Gemini AI API
Core Extensions:	Imagick, Tesseract OCR OCR processing engine

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1.1. Executive Summary

StudyShare is a state-of-the-art educational resource sharing platform designed to optimize knowledge distribution between administrative staff, faculty lecturers, and collegiate students. It offers a secure, highly accessible repository for academic notes, lecture slides, and study guides with embedded automated indexing and artificial intelligence enhancements.

At its core, StudyShare implements an intelligent OCR pipeline that automatically detects and digitizes handwritten or scanned PDF documents, uploading them directly to a securely bound Google Drive repository. The platform features an AI-driven Study Companion (utilizing the Gemini and Groq Large Language Model APIs) that extracts study points, creates structural summaries, and builds interactive practice flashcards and multiple-choice quizzes for students.

Recently, StudyShare underwent a major functional extension with the implementation of the 'Lecturer Subject Assignment System'. This feature provides academic administrators with a centralized control dashboard to assign one or multiple subjects to lecturers dynamically, based on study year, semester, and academic year partitions. This review outlines the complete technical specifications, underlying database designs, code structure, and logic pathways of the StudyShare codebase.

RELEASE NOTE: This document is generated programmatically following the completion of the Lecturer Subject Assignment System implementation, serving as the official system handbook.

2. System Architecture & Technical Stack

StudyShare is designed under a decoupled Client-Server architecture pattern, prioritizing maximum scalability, secure server operations, and rapid client interaction. The application logic runs on a standard PHP server (XAMPP/Apache) backstopped by a relational MySQL database server. The client layer utilizes modern semantic HTML5, highly responsive customized Vanilla CSS layouts, and micro-interactive JavaScript layers.

2.1 Technical Stack Breakdown

The system architecture integrates native technologies and external cloud integrations to construct an end-to-end learning environment:

- 1. Core Application Engine:** PHP 8.2 (Structured OOP/Procedural Hybrid) for server execution.
- 2. Relational Database:** MySQL 8.0, managed via highly optimized PDO (PHP Data Objects).
- 3. Cloud Storage Layer:** Google Drive API v3, managing high-volume file hosting dynamically.
- 4. Artificial Intelligence:** Gemini & Groq API LLM processing for smart study helper generation.
- 5. Image Processing:** Tesseract OCR & Imagick extension for scanned document digitization.

2.2 High-Level Integration Architecture

The server utilizes two dedicated folder IDs in Google Drive ('DRIVE_FOLDER_NORMAL' and 'DRIVE_FOLDER_SCANNED') to separate standard text-based files from OCR-processed image scans. High-speed database indexes optimize lookup speed on heavily queried relationship intersections. Authenticated user sessions are strictly isolated into distinct scopes: Admin, Lecturer, and Student.

ENVIRONMENT ISOLATION: External API configurations, credentials, and environmental specific paths are strictly isolated inside the root-level secure .env configuration file, which is loaded at bootstrap through standard parse_ini_file structures.

3. Core Data Model & Database Mapping (Part I)

StudyShare relies on a structured MySQL schema that ensures complete referential integrity. All key-value relationships are tied through foreign keys with standard cascading deletions. This prevents orphaned records when a note, lecturer, or student is removed.

Database Table: admin

Column Name	Data Type & Keys	Description & Logical Scope
id	INT PRIMARY KEY AUTO_INCREMENT	Unique identifier for administrative accounts.
username	VARCHAR(100) UNIQUE NOT NULL	Unique login name for administrators.
password	VARCHAR(255) NOT NULL	Secure MD5 hashed password for administrative access.
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP	Automatic record generation timestamp.

Database Table: students

Column Name	Data Type & Keys	Description & Logical Scope
roll_no	VARCHAR(50) PRIMARY KEY	Unique Student roll number (e.g. BCA25060).
name	VARCHAR(100) NOT NULL	Official student full name.
dob	DATE NOT NULL	Student date of birth. Serves as credentials (YYYY-MM-DD).
department	VARCHAR(50) NOT NULL	Associated academic department (e.g. BCA, BBA).
semester	INT NOT NULL	Student current academic semester (e.g., 2, 4, 6).
profile_image	VARCHAR(255) NULL	Filename of uploaded student avatar image file.

Database Table: lecturers

Column Name	Data Type & Keys	Description & Logical Scope
id	INT PRIMARY KEY AUTO_INCREMENT	Unique identifier for lecturer accounts.
username	VARCHAR(100) UNIQUE NOT NULL	Lecturer login roll number/username.
password	VARCHAR(255) NOT NULL	MD5 hashed password for account verification.
name	VARCHAR(100) NOT NULL	Official full name of the academic faculty lecturer.
email	VARCHAR(100) NULL	Official email address for correspondence.
department	VARCHAR(50) NOT NULL	Associated teaching department (e.g. BCA).
subject	VARCHAR(100) NULL	Legacy single subject string assigned to the lecturer.

4. Core Data Model & Database Mapping (Part II)

Database Table: lecturer_subject_assignments

Column Name	Data Type & Keys	Description & Logical Scope
id	INT PRIMARY KEY AUTO_INCREMENT	Unique Identifier for the assignment record.
lecturer_id	INT NOT NULL (FOREIGN KEY)	References lecturers.id. Cascades on deletion.
subject_id	INT NOT NULL (FOREIGN KEY)	References subjects.id. Cascades on deletion.
subject_name	VARCHAR(100) NOT NULL	Cached subject name for quick display lookups.
department	VARCHAR(50) NOT NULL	Department scoping of this specific assignment.
year	VARCHAR(20) NOT NULL	Target student academic year (e.g., 1st, 2nd, 3rd Year).
semester	INT NOT NULL	Target student semester (e.g. 2, 4, 6).
academic_year	VARCHAR(20) NOT NULL	Academic calendar year scope (e.g. 2025-2026).
status	TINYINT DEFAULT 1	Flag to toggle assignment active (1) or inactive (0).

Database Table: notes

Column Name	Data Type & Keys	Description & Logical Scope
id	INT PRIMARY KEY AUTO_INCREMENT	Unique Identifier for academic notes.
subject_id	INT NULL (FOREIGN KEY)	References subjects.id. Set Null on deletion.
lecturer_id	INT NULL (FOREIGN KEY)	References lecturers.id. Set Null on deletion.
description	TEXT NOT NULL	Lecturer input describing notes content.
normal_file_path	VARCHAR(255) NULL	Direct view Google Drive URL for standard PDF text notes.
scanned_file_path	VARCHAR(255) NULL	Direct view Google Drive URL for scanned/OCR processed PDFs.
ocr_text	LONGTEXT NULL	Text data extracted dynamically via Tesseract OCR engine.
is_scanned	TINYINT(1) DEFAULT 0	Boolean flag identifying whether notes require OCR processing.
likes	INT DEFAULT 0	Cache count of total likes gathered from students.

Database Table: smart_notes

Column Name	Data Type & Keys	Description & Logical Scope
id	INT PRIMARY KEY AUTO_INCREMENT	Unique Identifier for AI study guide companion records.
note_id	INT NOT NULL (FOREIGN KEY)	References notes.id. Cascades on parent note deletion.
summary	TEXT NOT NULL	AI generated bulleted summary of notes text.
key_points	TEXT NOT NULL	AI extracted critical learning point takeaways.
flashcards	TEXT NOT NULL	JSON structured questions and answers for study cards.

5. System Workflows & Sequential Pipeline Flows

5.1 Student Login & Authorization Pipeline

To guarantee frictionless accessibility for students in lab environments, authentication bypasses standard alphanumeric passwords in favor of an verified validation query using the student's Name and Date of Birth (DOB) as key combinations:

1. Student inputs Name and selects Date of Birth in the HTML interface.
2. The system executes a secure prepared SQL query selecting matching roll_no, department, and semester.
3. Upon successful validation, the system starts a secure session, caching the student's information (roll_no, department, and current semester) to scope note visibility correctly.

5.2 Smart Note Upload and OCR Process Flow

The system includes an automated processing pipeline that ensures files are indexed, scanned, OCR-analyzed, and offloaded to secure Google Drive folders:

1. A lecturer or admin uploads a lecture PDF. The system creates a temporary local file copy.
2. The OCR engine calls `extractTextFromPdf()` utilizing `pdftotext`. If the characters extracted are less than 100, the file is tagged as `is_scanned = 1`.
3. For scanned files, `Imagick` splits the PDF pages into PNG images at 300 DPI. `Tesseract OCR` parses the images to extract the raw text content (`$ocrText`).
4. Google Drive API (`DriveHelper`) is initiated. Standard digital notes are written to `DRIVE_FOLDER_NORMAL`, while scanned notes are written to `DRIVE_FOLDER_SCANNED`.
5. The file permissions are made public read, the Drive view link is fetched, and the database record is created. The local file is deleted to preserve web server disk space.

5.3 AI-driven Study Companion Pipeline

Once note text is extracted, the student can toggle 'Study Mode':

1. The system reads the notes table records (either raw digital text or `ocr_text`).
2. If no smart summary exists, it forms a structured prompt and sends it via `cURL` to the Gemini/Groq APIs.
3. The AI generates a formatted study package (detailed summaries, key takeaways, and flashcards). The JSON is parsed, validated, and stored in the database's `smart_notes` table for instant student study access.

CLOUD STORAGE HIGHLIGHT: This automated offloading pipeline means the local server's file system is never clogged with large documents, while Google Drive takes care of standard, robust file delivery.

6. Detailed Review of the Lecturer Subject Assignment System

The Lecturer Subject Assignment System is a major module designed to replace rigid, hardcoded, one-to-one lecturer-subject configurations with a multi-subject, semester-scoped, calendar-driven system. This is done without altering existing tables, ensuring perfect backward compatibility.

6.1 High-Level Functional Specifications

- Multi-Subject Assignment: Admin can assign multiple subjects to a single lecturer concurrently utilizing a grid of checkboxes with Search-to-Filter and Select-All utilities.
- Academic Year & Year/Sem Isolation: Assignments are divided by year (1st, 2nd, 3rd) and semester (1 to 6), allowing a lecturer to teach different subjects simultaneously or across academic semesters.
- Duplicate Prevention: The server performs a pre-submission duplicate check, ensuring that the same lecturer cannot be assigned the same subject in the same semester and academic year twice.
- Active/Inactive Status Toggle: An assignment can be set active or inactive. This allows the administrator to pause a lecturer's access to a course without deleting historical records.

6.2 DB Schema Integration

A new table, `lecturer_subject_assignments`, is added with foreign key constraints targeting the parent lecturers and subjects tables. It caches subject names and departments to minimize database JOIN operations during page rendering, greatly improving load speeds.

6.3 Lecturer Portal & Student Portal Integration Flow

1. Note Uploads in Lecturer Portal: Previously, lecturers could only upload notes to a single, hardcoded subject. Now, the dashboard queries `lecturer_subject_assignments` where `status = 1`, and populates a dynamic dropdown. Lecturers choose which assigned course they are uploading notes for.
2. Note Filtering in Student Portal: The student portal queries the `lecturer_subject_assignments` table to see which lecturers are teaching their semester's subjects. The students then view notes organized cleanly under these exact subjects.

AUDIT TRAIL INTEGRATION: The subject assignment system is fully tracked. Any assignment action (creation, toggle, deletion) is logged in the `activity_log` table for administrative audit trails.

7. Technical Map of Codebase File & Directory Structure

The StudyShare codebase is clean, structured, and modular. Code elements are neatly divided by target actors (admin, lecture, student, user) and shared helper functions (common):

studyshare/

- admin/	
- activity_log.php	- Renders activity logs, provides stats charts
- assign_subjects.php	- Administrative subject-to-lecturer dashboard
- dashboard.php	- Admin home, displays metrics and user management
- manage_notes.php	- Notes edit/delete options for administrators
- manage_students.php	- Adds, edits, or deletes student profiles
- upload_notes.php	- Allows admins to upload notes directly
- login.php	- Admin secure entry page
- assets/	
- css/	- Admin, lecturer, student and login stylesheets
- js/	- Client-side interactivity scripts
- common/	
- config.php	- Key variables, directory definitions, API limits
- db.php	- PDO connection manager, smart-notes table creator
- drive_helper.php	- Google Drive API upload, sharing, and deletion library
- functions.php	- Authentication check, file validation, activity logs
- ocr_helper.php	- Tesseract, pdftotext, and Imagick PDF processing wrapper
- subjects.php	- Subject lists, compulsory vs language-specific rules
- database/	
- studyshare.sql	- Relational database structure and default dataset
- lecture/	
- dashboard.php	- Lecturer panel. Uploads notes to assigned subjects
- profile.php	- Manages lecturer details and profile images
- login.php	- Lecturer entry page
- student/	
- index.php	- Legacy entry page for students
- uploads/	
- notes/	- Local note cache (cleared after Drive upload)
- scanned/	- Local OCR image-split cache (cleared after OCR)
- profile/	- Uploaded student profile avatars
- user/	
- dashboard.php	- Main Student UI. Renders dynamic subject blocks
- subject-notes.php	- Lists notes for selected subject, triggers smart mode
- process_note.php	- Gemini API controller for summaries and flashcards
- generate_quiz.php	- AI multiple choice practice quiz manager
- login.php	- Student Name & DOB credential entry
- .env	- Encrypted storage for API keys and DB passwords
- composer.json	- Standard dependency list (requires FPDF and Google Client)

CODEBASE MODULARITY: This layout isolates system credentials into the root `.env`, backend operations in `common/`, and views into distinct folders, ensuring a strict Model-View-Controller design pattern.

8. Security Specifications, Performance & Best Practices

8.1 SQL Injection Protection

All SQL statements inside the StudyShare codebase are executed utilizing PHP Data Objects (PDO) prepared statements. No variables are directly concatenated into SQL strings. This completely neutralizes SQL Injection (SQLi) attacks. Parameter binding is used to safely process student DOBs, roll numbers, search strings, and lecturer subject assignment records.

8.2 File Upload Validation Sandbox

To block execution of malicious PHP scripts via note uploads, a multi-tier sandbox validation is implemented:

1. Strict File Extension Matching: The system uses a strict whitelist of file extensions ('pdf', 'doc', 'docx') validated on the server via `pathinfo()`.
2. MIME-Type Whitelisting: The browser-provided mime-type is ignored, and the server validates the actual file contents against a whitelist of acceptable formats.
3. Storage Isolation: Note files are quickly processed and offloaded to Google Drive. No PDF files are stored permanently on the local web server, ensuring no local execution is possible.

8.3 High-Load Performance Optimizations

- Database Indexing: Crucial indexes are created on notes(department, semester), students(roll_no), and likes(note_id) to avoid table-scans on high traffic.
- Image Resolution Tuning: In Imagick PDF-to-Image processing, the resolution is set at 300 DPI. This is the optimal threshold for clean OCR recognition without hogging CPU/RAM.
- Cloud API Offloading: Storing files on Google Drive keeps network bandwidth load low on the local server.

SECURE ROUTING PROTECTION: The session validation files (`auth_admin.php`, `auth_lecturer.php`, `auth_user.php`) run at the top of every restricted file to verify active, unexpired sessions, preventing illegal access to dashboard metrics.

9. Verification & Concluding Summary

9.1 Automated and Manual Test Plans

Verification of the implemented features was executed across multiple dimensions to confirm operational robustness:

- 1. Database Schema Integrity:** Verified table relationships, index lookup speeds, and cascades.
- 2. Admin Assignment Suite:** Tested multi-subject checkboxes, duplicate assignments block, and filters.
- 3. Lecturer Dropdowns:** Verified notes upload subject lists matched dynamic assigned scopes.
- 4. Student Grid Rendering:** Checked that subject folders matched the lecturer assigned subjects.

9.2 Concluding Summary

The StudyShare platform successfully combines local server reliability with external cloud integrations. The addition of the 'Lecturer Subject Assignment System' gives administrators complete control over teaching assignments without breaking any previous system behavior. In addition, the automated OCR processing and LLM-driven companion features transform StudyShare from a simple storage system into an intelligent, active learning partner for students.

The codebase continues to grow in a modular and organized way. Adherence to strict MVC design, security practices (SQLi prevention, sandboxing, session isolation), and performance configurations makes StudyShare extremely reliable, secure, and ready for deployment.

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